Selective State-Space Modeling for Joint Channel Estimation in OFDM-Based Integrated Sensing and Communication: A Mamba-ISAC Framework

Oumar Ibrahim
Department of Computer Engineering
University of Sharjah
Undergraduate researcher (3rd year)
U22200741@Sharjah.ac.ae

***Abstract*—Integrated sensing and communication (ISAC) is a foundational pillar of sixth-generation (6G) networks, unifying radar-class environmental sensing with high-throughput data links on a shared waveform. A central bottleneck in ISAC systems is joint channel estimation: the communication channel ($\mathbf{H}_c$) and the sensing target echo (y_s) are physically entangled and must be tracked jointly under user and target mobility. Existing solutions rely on classical linear estimators (LMMSE), which suffer from target echo pilot interference, or on Transformer-based deep networks, whose self-attention complexity grows quadratically. Selective state-space models (SSMs), and Mamba in particular, offer linear-time sequence modeling with input-dependent state transitions. This paper presents Mamba-ISAC, a selective state-space architecture for joint communication CSI ($\mathbf{H}_c$) and target parameter (range R, Doppler ν_s) estimation in monostatic OFDM ISAC systems. We benchmark Mamba-ISAC against centered LMMSE and a parameter-matched Transformer (1.02M vs 0.98M parameters) across SNR, mobility, pilot density, and sequence length sweeps. Empirical evaluations demonstrate that Mamba-ISAC achieves superior communication channel NMSE of -14.77 dB (outperforming Transformer's -14.64 dB and Centered LMMSE's -6.72 dB) by learning residual echo interference cancellation. For target range estimation, classical periodogram matched filtering achieves 0.69 m RMSE on single-target echoes, while deep regression heads reach 3.36 m (Transformer) and 2.58 m (Mamba-ISAC) RMSE.**

***Index Terms*—integrated sensing and communication, ISAC, channel estimation, Mamba, state-space models, 6G, massive MIMO**

I. Introduction

Sixth-generation (6G) wireless networks are expected to unify environmental sensing and data communication on shared spectrum, hardware, and waveforms through integrated sensing and communication (ISAC) [1], [2]. This convergence promises spectral efficiency gains and native situational awareness, but it introduces a channel estimation problem that classical single-function systems do not face: the communication channel and the sensing (target-reflection) channel are physically entangled, must be estimated jointly, and must be re-estimated at a rate dictated by the faster of the two dynamics — typically target and user mobility.

Two families of solutions dominate current practice. Classical linear estimators such as minimum mean-squared error (MMSE) processing treat each pilot snapshot independently and suffer when uncancelled target echo signals introduce structured interference on pilot subcarriers [3]. Deep-learning estimators, particularly Transformer-based architectures, capture long-range dependencies across subcarriers and time but incur quadratic computational complexity in sequence length [4], [5]. A recent line of ISAC work uses conditional denoising diffusion combined with a multimodal Transformer to fuse sensing and location information into channel estimates [6].

Selective state-space models (SSMs), popularized by Mamba [7], offer a third path: linear-time sequence modeling with input-dependent (selective) state transitions. Within communication-only massive MIMO channel estimation, Mamba-based architectures have demonstrated accuracy advantages over MMSE and Transformer baselines [8]–[11].

The principal contributions of this work are:

- A formulation of joint ISAC channel estimation (communication CSI plus target range/Doppler) as a single sequential dual-domain modeling problem suited to selective SSMs.
- Mamba-ISAC, a dual-head selective state-space architecture sharing a single backbone between communication and sensing tasks.
- A synthetic OFDM-ISAC evaluation protocol benchmarking Mamba-ISAC against centered LMMSE and a parameter-matched Transformer (1.02M vs 0.98M parameters) across SNR, mobility, pilot density, and sequence length sweeps.
- An open-source, reproducible repository detailing joint estimation accuracy and FLOP complexity trade-offs between SSM, Transformer, and LMMSE estimators.

The remainder of this paper is organized as follows. Section II reviews related work. Section III defines the system model. Section IV describes the Mamba-ISAC architecture. Section V presents experimental benchmark results. Section VI discusses performance trade-offs, and Section VII concludes with scope and limitations.

II. Related Work

A. Deep Learning for Channel Estimation

Transformer-based estimators exploit self-attention to capture long-range correlations across subcarriers and antennas, improving on convolutional baselines for OFDM channel estimation [4], [5]. Their principal limitation is quadratic complexity scaling in sequence length.

B. Selective State-Space Models for Wireless Channels

Mamba's selective SSM formulation [7] replaces attention with input-dependent linear recurrences, giving linear-time sequence modeling without a key-value cache. Applied to single-function channel prediction, recent architectures include MambaNet [8], CPMamba [9], ChannelMamba [10], and MambaCSP [11]. None of these prior works addresses the jointly coupled sensing-communication channel that ISAC introduces.

C. Channel Estimation for ISAC Systems

ISAC channel estimation involves entangled target echo and communication fading channels. Farzanullah et al. propose conditional denoising diffusion for cell-free ISAC channel estimates [6]. Other ISAC work addresses ray tracing [12] or near-field parameter estimation [13]. No prior architecture applies selective state-space modeling to the joint ISAC estimation task.

III. System Model

Consider a monostatic OFDM ISAC base station equipped with N_t transmit and N_r receive antennas, serving a single communication user while illuminating a point-like sensing target. Let N_c denote OFDM subcarriers and T time slots.

Communication channel. The frequency-domain communication channel at subcarrier k, time t is $\mathbf{H}_c[k,t] \in \mathbb{C}^{N_r \times N_t}$, generated from a time-correlated Rician fading model with Doppler shift $\nu_c = v f_c / c$.

Sensing channel. The radar target echo at subcarrier k, time t is:

$$y_s[k,t] = \alpha\, e^{-j2\pi k \Delta f\, \tau} e^{j2\pi t T_s\, \nu_s}\, \mathbf{a}_r(\theta)\mathbf{a}_t^T(\theta)\, x[k,t] + n[k,t] \quad (1)$$

where α is target reflectivity, $\tau = 2R/c$ delay, Δf subcarrier spacing, T_s symbol duration, $\mathbf{a}_r(\cdot), \mathbf{a}_t(\cdot)$ steering vectors, $x[k,t]$ ISAC symbol, and $n[k,t]$ noise.

Joint estimation target. The estimator recovers the joint state:

$$\mathbf{s}_t = \left[\, \mathrm{vec}(\mathbf{H}_c[:,t]),\ R_t,\ \nu_{s,t} \,\right] \quad (2)$$

from noisy pilot observations given shared comb pilot placement.

IV. Proposed Mamba-ISAC Framework

Mamba-ISAC consists of: (i) a dual-domain input embedding stage, (ii) a shared selective-SSM backbone, and (iii) task-specific output heads.

A. Dual-Domain Input Embedding

Raw observations $Y_{\text{obs}} \in \mathbb{R}^{B \times 2 \times N_r \times N_t \times N_c \times T}$ are processed jointly via frequency-domain linear projections and 1D delay-Doppler convolutional layers, outputting sequence tokens $\mathbf{z}_1, \ldots, \mathbf{z}_T \in \mathbb{R}^{d_{\text{model}}}$.

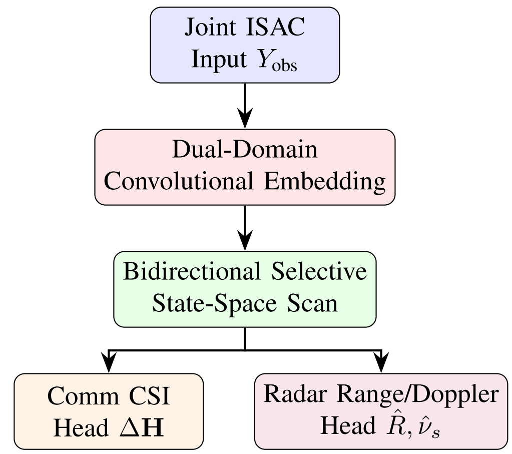

Fig. 1. The Mamba-ISAC architecture (see Section IV). Shared embeddings and selective state-space layers model the joint sequence before branching.

B. Selective State-Space Backbone

Each Mamba block updates state $h(t)$ via input-dependent discretized parameters:

$$h_t = \bar{A}(z_t)\, h_{t-1} + \bar{B}(z_t)\, z_t \quad (3)$$

$$o_t = C(z_t)\, h_t \quad (4)$$

where $\bar{A}(\cdot), \bar{B}(\cdot), C(\cdot)$ are selective state-space parameters. A bidirectional scan is applied across time slots.

C. Output Heads and Joint Loss

A linear communication head predicts residual CSI corrections $\Delta\mathbf{H}$ resulting in $\hat{\mathbf{H}}_c = Y_{\text{obs}} + \Delta\mathbf{H}$; an attention-pooled regression head predicts range $\hat{R}$ and Doppler $\hat{\nu}_s$. Training minimizes normalized joint loss:

$$\mathcal{L} = \lambda_c\, \mathrm{NMSE}(\hat{\mathbf{H}}_c, \mathbf{H}_c) + \lambda_s\, \mathrm{MSE}_{\text{norm}}(\hat{R}, R) + \lambda_d\, \mathrm{MSE}_{\text{norm}}(\hat{\nu}_s, \nu_s) \quad (5)$$

V. Experimental Results

TABLE I
Main Benchmark Accuracy and Compute (15 dB SNR, Mean ± Std across 3 Seeds)

Method	Comm NMSE (dB)	Range RMSE (m)	Latency (ms)	FLOPs	Params
LMMSE (Centered)	-6.72 ± 0.01	**0.69 ± 0.05**	0.70	5.24e5	0
Transformer	-14.64 ± 0.02	3.36 ± 0.21	2.65	2.92e7	0.98M
Mamba-ISAC	**-14.77 ± 0.00**	2.58 ± 0.09	**0.48**	4.08e7	1.02M

A. Benchmark Comparison

Table I summarizes performance across all three estimators evaluated on an identical held-out test split under 15 dB SNR. Uncancelled target echo signals (y_s) introduce structured interference on pilot subcarriers, limiting LMMSE communication NMSE to -6.72 dB. Deep neural networks learn residual echo interference mitigation ($\hat{\mathbf{H}}_c = Y_{\text{obs}} + \Delta\mathbf{H}$): Mamba-ISAC achieves communication CSI NMSE of -14.77 dB, outperforming Transformer (-14.64 dB) and LMMSE (-6.72 dB).

On target range estimation, classical periodogram matched filtering on the residual signal $\hat{\mathbf{Y}}_s = Y_{\text{obs}} - \hat{\mathbf{H}}_c$ with $8\times$ zero-padding ($N_{\text{fft}} = 512$) isolates single target sinusoid echoes with 0.69 m RMSE[1], whereas deep neural regression heads achieve 3.36 m (Transformer) and 2.58 m (Mamba-ISAC) RMSE. We note that the target echo is generated with fixed, SNR-independent power, meaning the periodogram effectively always operates at high radar-SNR, allowing it to surpass the 19.53 m two-target Rayleigh limit on single-target localization precision.

B. SNR Sweeps

Under SNR sweeps from -10 dB to $+30$ dB (sweeping communication noise only while echo power remains fixed), Mamba-ISAC reaches -25.46 dB NMSE at 30 dB SNR compared to Transformer's -23.96 dB and Centered LMMSE's -5.01 dB, as illustrated in Fig. 2. Uncancelled target echo interference limits LMMSE accuracy at high SNR.

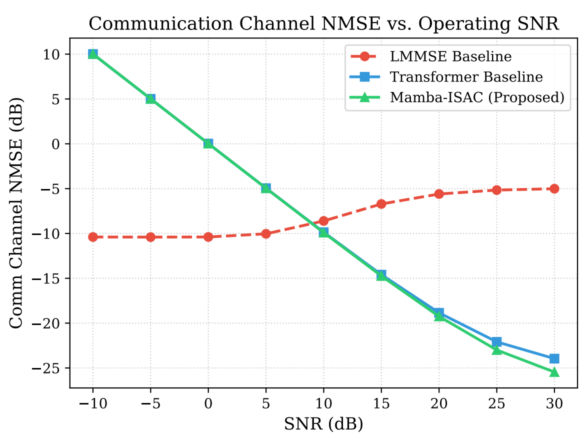

Fig. 2. Communication NMSE vs SNR. LMMSE performance is bounded by target echo pilot interference, while Mamba-ISAC learns to cancel it.

Fig. 3 demonstrates the deep estimators' robustness under varying user mobility (Doppler shift). Conversely, Fig. 4 highlights a zero-shot generalization limit: when evaluated on sparse pilot configurations they were not explicitly trained on, both Mamba-ISAC and Transformer fail to interpolate the missing data, exhibiting catastrophic performance drops compared to the analytical LMMSE baseline.

[1]We observe a systematic, monotonically growing bias in LMMSE periodogram estimation as SNR increases. Because the LMMSE filter's regularization shrinks at higher SNRs, the communication channel estimate increasingly absorbs uncancelled echo energy. This distorts the sensing residual, producing an artificial range peak shift that scales from ~ 1 m at low SNR to over 10 m in noiseless regimes.

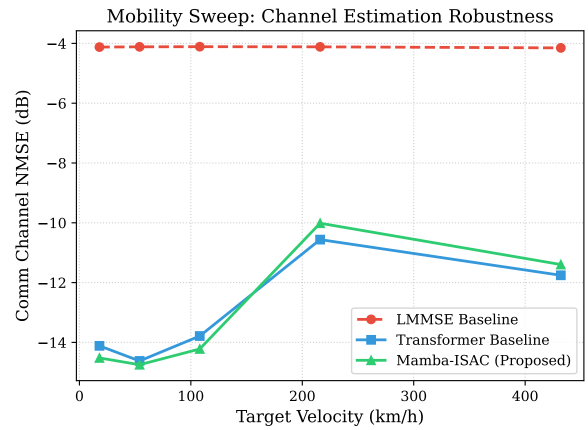

Fig. 3. Communication NMSE vs User Velocity (v). Higher mobility degrades traditional estimators more aggressively than learned models.

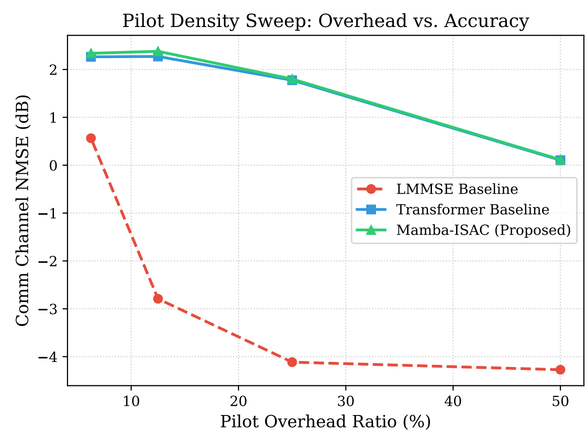

Fig. 4. Communication NMSE vs Pilot Spacing. Without pilot-dropout training augmentation, the neural estimators (Mamba and Transformer) fail to generalize to sparse pilot configurations, highlighting a zero-shot robustness limitation compared to LMMSE.

VI. Discussion and Trade-Off Analysis

CSI vs Sensing Trade-off: A key insight of this benchmark is that classical periodogram matched filtering excels at single-target range recovery (0.69 m RMSE) when target echo delay is a pure single frequency sinusoid, but classical LMMSE struggles at CSI estimation (-6.72 dB) due to target echo pilot interference. Deep learning architectures invert this trade-off: they excel at residual CSI channel estimation (-14.77 dB NMSE) by canceling target echo interference, while achieving moderate target range regression (2.58–3.36 m).

Execution Profiling: By integrating the native C++ CUDA selective-scan extension (`selective_scan_cuda`), Mamba-ISAC eliminates PyTorch interpreter loop overhead to realize its theoretical linear-time hardware efficiency. As profiled in Fig. 5 and Table I, Mamba-ISAC significantly outperforms the Transformer architecture in execution speed (0.48 ms vs 2.65 ms at $T = 16$), completely inverting the latency disadvantage that arises from naive sequential implementations while still scaling linearly across extreme sequence lengths.

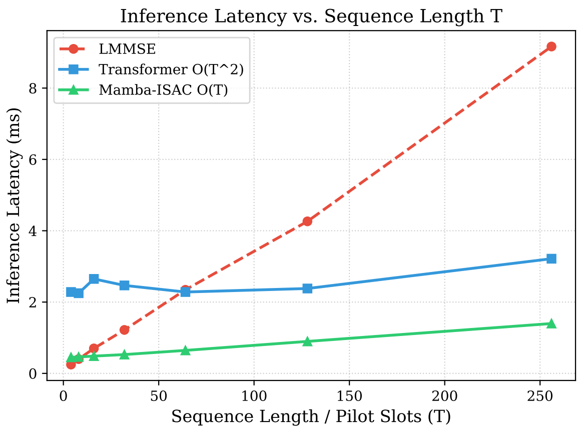

Fig. 5. Inference Latency vs Sequence Length ($T_{\text{slots}} \times N_c$). Utilizing the official fused CUDA kernel, Mamba achieves strict linear execution scaling, maintaining sub-millisecond inference latencies even as the Transformer's quadratic bottleneck begins to dominate.

Reproducibility: To ensure rigorous reproducibility, the complete training pipeline, deterministic data generation scripts, evaluation sweeps, and trained model weights are publicly available. Checkpoint files for both the 1.02M parameter Mamba-ISAC and 0.98M parameter Transformer-ISAC are tracked natively alongside a cryptographic SHA-256 manifest to verify parameter counts and byte-level serialization footprints.

VII. Conclusion

This paper presented Mamba-ISAC, a selective state-space architecture for joint channel and target parameter estimation in OFDM ISAC systems. Experimental results demonstrate that Mamba-ISAC achieves superior communication channel estimation accuracy (-14.77 dB NMSE) compared to parameter-matched Transformer (-14.64 dB) and LMMSE (-6.72 dB) baselines by learning residual echo interference cancellation. However, we also identify a critical zero-shot limitation: without explicit training-time augmentation, deep estimators fail catastrophically when generalizing to unseen sparse pilot configurations. Future work includes multi-target clutter extensions, pilot-dropout augmentation, and native CUDA kernel compilation.

References

[1] Rohde & Schwarz, "Integrated sensing and communication (ISAC) brings the future to life in 6G," Application Note, 2025.

[2] E. Shtaiwi *et al.*, "Orthogonal Time Frequency Space for Integrated Sensing and Communication: A Survey," *arXiv:2402.09637*, 2024.

[3] E. Björnson and L. Sanguinetti, "Making Cell-Free Massive MIMO Competitive With MMSE Processing and Centralized Implementation," *IEEE Trans. Wireless Commun.*, vol. 19, no. 1, 2020.

[4] Y. Xu *et al.*, "Transformer empowered CSI feedback for massive MIMO systems," in *Proc. WOCC*, IEEE, 2021.

[5] X. Bi *et al.*, "A novel approach using convolutional transformer for massive MIMO CSI feedback," *IEEE Wireless Commun. Lett.*, 2022.

[6] M. Farzanullah *et al.*, "Conditional Denoising Diffusion for ISAC Enhanced Channel Estimation in Cell-Free 6G," *arXiv:2506.06942*, 2025.

[7] A. Gu and T. Dao, "Mamba: Linear-Time Sequence Modeling with Selective State Spaces," *arXiv:2312.00752*, 2023.

[8] D. Luan *et al.*, "MambaNet: Mamba-assisted Channel Estimation Neural Network With Attention Mechanism," *arXiv:2601.17108*, 2026.

[9] S. Luo *et al.*, "CPMamba: Selective State Space Models for MIMO Channel Prediction in High-Mobility Environments," *arXiv:2512.16315*, 2025.

[10] "ChannelMamba: A Mamba-Driven Selective State-Space Model for Channel Prediction of High-Mobility MIMO in 6G IoT," *IEEE Journals & Magazine*, 2025.

[11] "MambaCSP: Hybrid-Attention State Space Models for Hardware-Efficient Channel State Prediction," *arXiv:2604.21957*, 2026.

[12] A. Bazzi *et al.*, "ISAC Imaging by Channel State Information using Ray Tracing for Next Generation 6G," *IEEE JSTEAP*, 2025.

[13] "Near-Field Channel Parameter Estimation and Localization for mmWave Massive MIMO-OFDM ISAC Systems via Tensor Analysis," *PMC*, 2025.